

The polylog-bounded class and its closure under power steps

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Abstract

To analyse the general tower of unit 55 when the exponents are sorted in non-increasing order, its levels must be tracked through a class closed under the two kinds of power step that occur. This unit introduces `PolyLogBounded` $G \in C^j$ (measurable, $0 \leq G \leq C_0 y^\delta$ on $(0, 1]$, $G \leq C_1(1 + \log y)^j$ on $[1, \infty)$) and proves: the base level F_{q-1} is in the class with $j = 0$; the logarithmic step $\int_0^y G(x)/x dx$ raises j by one; a convergent step $\int_0^y x^t G(x) dx$ with $t < -1$ is bounded by its limit and is a `LogBase` with tail rate $(-t - 1)/2$. Zero `sorry/axiom`.

1 Setting

The only obstacle to treating polylog-bounded functions like bounded ones is the logarithm in the growth bound. It is absorbed by $(1 + \log x)^j \leq (1 + 1/\varepsilon)^j x^{j\varepsilon}$ for $x \geq 1$ (from $\log x \leq x^\varepsilon/\varepsilon$), with $\varepsilon = (-t - 1)/(2(j + 1))$ chosen so that $x^t(1 + \log x)^j \leq Cx^{(t-1)/2}$, still integrable at infinity with exponent $(t - 1)/2 < -1$ and tail $\int_y^\infty x^{(t-1)/2} = \frac{2}{-t-1}y^{-(t-1)/2}$.

2 The things formalised

```
structure PolyLogBounded (G :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (C C :  $\mathbb{R}$ ) (j :  $\mathbb{N}$ ) : Prop where
  measurable : Measurable G
  nonneg :  $y, 0 \leq G y$ 
  _pos :  $0 < y \rightarrow y^{-1} \rightarrow G y \leq C * y^{-j}$ 
  le_pow :  $y, 0 < y \rightarrow y^{-1} \rightarrow G y \leq C * y^{-j}$ 
  le_log :  $y, 1 \leq y \rightarrow G y \leq C * (1 + \text{Real.log } y)^j$ 
theorem one_add_log_pow_le_rpow (j :  $\mathbb{N}$ ) (x :  $\mathbb{R}$ ) (h :  $0 < x$ ) (hx :  $1 \leq x$ ) :
   $(1 + \text{Real.log } x)^j \leq (1 + 1/x)^j * x^{((j : \mathbb{N}) * j)}$ 
theorem PolyLogBounded.powStep_neg_one (hG : PolyLogBounded G C C j) :
  PolyLogBounded (powStep (-1) G) (C / 2) (C / 2 + C / ((j :  $\mathbb{N}$ ) + 1)) (j + 1)
theorem PolyLogBounded.powStep_of_lt (hG : PolyLogBounded G C C j) (t :  $\mathbb{R}$ ) (ht :  $t < -1$ ) (ht' :  $-1 < t + 1$ ) :
  PolyLogBounded (powStep t G) (C / (t + 1)) (t + 1) (powLimit t G) 0
theorem PolyLogBounded.logBase_powStep_of_lt (hG : PolyLogBounded G C C j) (t :  $\mathbb{R}$ ) (ht :  $t < -1$ ) (ht' :  $-1 < t + 1$ ) :
  LogBase (powStep t G) (powLimit t G) (C * (1 + 1 / absorbEps t j)^j * (2 / (-t - 1)))
  (C / (t + 1)) (t + 1) ((-t - 1) / 2)
theorem weightedPrimitive_polyLog (a q :  $\mathbb{R}$ ) (h :  $0 < a$ ) (hq :  $0 < q$ ) :
  PolyLogBounded (weightedPrimitive a (q - 1)) (Real.exp (a * a^2 / 2) / q) q
  (weightedMass a (q - 1)) 0
```

3 Proof strategy

Integrability of $x^t G(x)$ on $(0, L]$ for $t + \delta > -1$ splits at 1: the near-0 bound $C_0 x^{t+\delta}$ and, on $(1, L]$, the constant $(L^t + 1)C_1(1 + \log L)^j$ (using $x^t \leq L^t + 1$ in both signs of t). Monotonicity of `powStep` (nonnegative integrand over a growing set) gives measurability. The near-0 bound propagates via $\int_0^y x^{t+\delta} = y^{t+\delta+1}/(t + \delta + 1)$. The logarithmic step uses unit 39's closed form $\int_1^y (1 + \log x)^j/x = ((1 + \log y)^{j+1} - 1)/(j + 1)$ and $(1 + \log y)^{j+1} \geq 1$ to absorb the $(0, 1]$ piece. The convergent step uses the absorption inequality, `integrableOn_Ioi_rpow_of_lt` and `integral_Ioi_rpow_of_lt`; positivity of the limit comes from positivity of the integrand on $(0, 1]$.

4 Roadblocks and resolutions

- `div_le_div_iff_of_pos` does not exist here; `div_le_div_iff` does.
- `gcongr` closed a goal entirely (including the side condition) so trailing bullets errored; replaced by an explicit `mul_le_mul_of_nonneg_left`.
- `positivity` cannot see $0 < \varepsilon$ for the defined `absorbEps`; bring the positivity lemma into context first.

5 What was Mathlib, what was new

Mathlib: `Real.log_le_rpow_div`, `Real.one_le_rpow`, `Real.rpow_le_rpow`, `Real.rpow_le_one_of_one_le_of_nonpos`, `Real.rpow_le_rpow_of_exponent_le`, `div_le_div_iff`. New: the class, both closure theorems, the `LogBase` certificate for convergent steps, and the base instance.

6 Lessons

Designing the class around exactly the two operations that will be applied (logarithmic and convergent steps) kept the invariant small; the third kind of step (divergent, $t > -1$) is avoided by sorting the coordinates, which is cheaper than tracking polynomial growth.

7 Outcome

All ingredients for the sorted-exponent tower are in place: the base is in the class, noncritical steps stay in the class, the first minimal step produces an admissible base, and the remaining minimal steps are `iterDivPrim`. The next unit assembles them into the general “only minimal exponents count” theorem for sorted exponent vectors.